COS 30049 Computing Technology Innovation

Project
Week4: Data Processing via Python (Lecture-02)

Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.



Objectives:

1. Understanding data structure (CSV, text, JSON)
2. Data Preprocessing

Everything here feeds the Data Processing section of Assignment 2 (A2)

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Data processing language

- Python (we will be using)
  - **Multi-Purpose Language:**

    Fully fledged, multi purpose programming language. Being widely used in various domains, including web development, automation, and data science.

  - **Integration:**

    Good for combining data wrangling activities into a larger pipeline of production or web development

  - **Libraries and Frameworks:**

    Good library support for scientific and machine learning extensions, has a rich ecosystem of machine learning libraries such as TensorFlow, PyTorch, etc. These libraries are well documented and supported by a large community.

  - **Community and Support:**

    Python has a large and active community, making it easier to find resources, tutorials, and support for machine learning projects.

- R (we will not be using)
  - More of a statistics focus.
  - R might have a steeper learning curve for those not familiar with statistical concepts.
  - R is generally slower than Python when handling large datasets.

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Open Data (A2)

Open data is publicly available for reuse. Open data can be accessed from a range of sources. High quality data can be found on government websites and institutional repositories.

- Government data and statistics in Australia
  - Australian Government Data
  - Victoria Government Data
  - Australian Bureau of Statistics
  - Closing the Gap Information Repository
- Institutional Repositories - consider the credibility before using
  - Google Dataset Search
  - HuggingFace
  - Kaggle

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Dataset Quality Review (A2)

Evaluate the quality of the dataset, just as you would evaluate any information you find, before you use it in your assignments or projects.

- **Authority**:
  - Who collected the data
  - Is it an educational institution, government or a reputable organisation?
  - If an individual has produced the dataset, are they associated with a reputable organisation or are they well–known in their field?
- **Coverage**:
  - Were enough samples taken to be representative of the total population or group being researched?
  - Is the timeframe relevant or up-to-date?

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Dataset Quality Review using python (A2)

Sometimes, to gain a deeper understanding of data quality, we need to use Python to examine the dataset's quality.

- We can use `pandas` package in python to do extensive data exploration
  - A basic descriptive statistics: `describe()` method for an overview of key statistical metrics.

```
In [37]: # Using Pandas to describe the 'Y2013' column - a column of numbers (integers)
         data['Y2013'].describe()

Out[37]: count     21477.000000
         mean        575.557480
         std        6218.379479
         min        -246.000000
         25%           0.000000
         50%           8.000000
         75%          90.000000
         max      489299.000000
         Name: Y2013, dtype: float64

In [35]: # Using Pandas to describe the 'Area' column - a column of strings
         data['Area'].describe()

Out[35]: count     21477
         unique      174
         top       Spain
         freq        150
         Name: Area, dtype: object
```

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Sample size (A2)

Decide how many responses or observations you need to have a good sample size. Larger sample sizes are more likely to allow you to draw accurate conclusions than smaller samples.

- Sample Size Formula (Simplified):

$$n = \frac{Z^2 \cdot p \cdot (1 - p)}{e^2}$$

  - n = required sample size
  - Z = Z-value (e.g., 1.96 for 95% confidence)
  - p = estimated proportion (e.g., 0.5 for maximum variability)
  - e = margin of error (0.01-0.1)

Normally 3000–4000

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Understand data structure (CSV, text, JSON) (A2)

Understanding CSV (Comma-Separated Values):

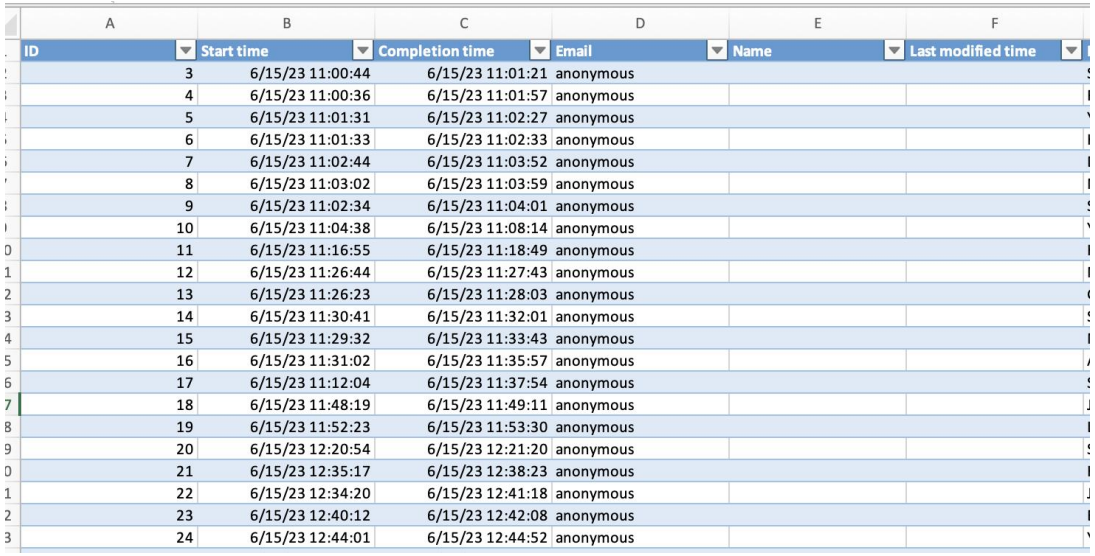
- **Definition**: A plain text format for storing tabular data where each line represents a row, and columns are separated by commas.
- Widespread Use in Spreadsheets:
  - Huge amounts of data lives in spreadsheets
    - Businesses
    - Hospitals
    - Education
    - …
- Accessible via popular tools:
  - Can by accessed by Microsoft (Excel), OpenOffice (Calc), Google Sheets
- Other Advantages of CSV:
  - Human-Readable Format: Easy to view and edit using any text editor.
  - Compatibility: Widely supported across different software and platforms.
  - Simplicity: Lacks the complexity of binary formats like XLS.
- Working with CSV in Python:
  - Popular Library: pandas
  - **Key Function:** read_csv()
    - Example: df = pandas.read_csv('file.csv')
    - Easily load CSV data into a DataFrame for analysis.

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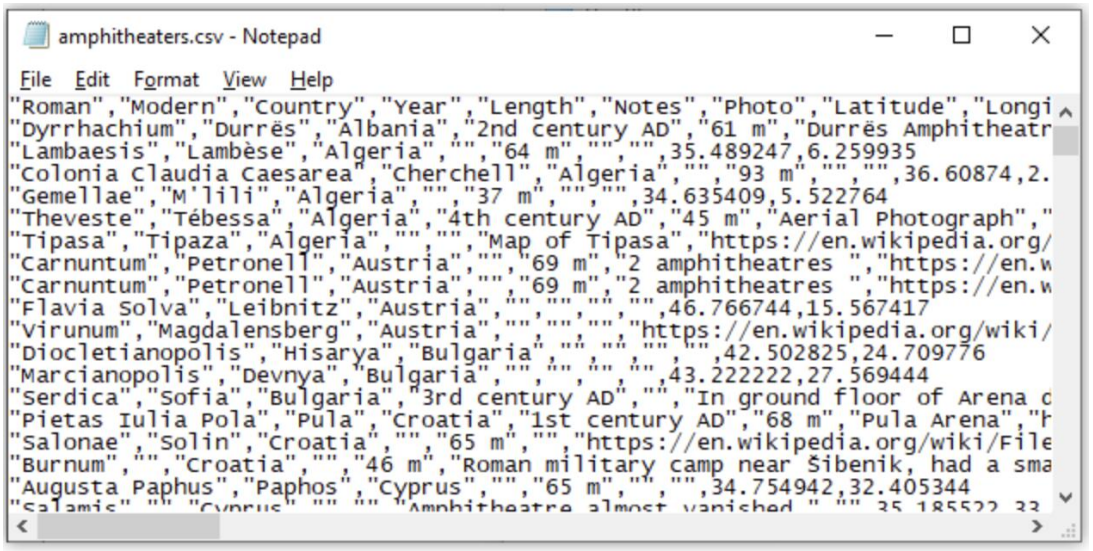
Understand data structure (CSV, text, JSON)

Example - Comma Separated Values (CSV) (A2)

- Spreadsheets
- Easy to use
- Structured
- Information in tabular format, simple but many entries



ID	Start time	Completion time	Email	Name	Last modified time
3	6/15/23 11:00:44	6/15/23 11:01:21	anonymous		
4	6/15/23 11:00:36	6/15/23 11:01:57	anonymous		
5	6/15/23 11:01:31	6/15/23 11:02:27	anonymous		
6	6/15/23 11:01:33	6/15/23 11:02:33	anonymous		
7	6/15/23 11:02:44	6/15/23 11:03:52	anonymous		
8	6/15/23 11:03:02	6/15/23 11:03:59	anonymous		
9	6/15/23 11:02:34	6/15/23 11:04:01	anonymous		
10	6/15/23 11:04:38	6/15/23 11:08:14	anonymous		
11	6/15/23 11:16:55	6/15/23 11:18:49	anonymous		
12	6/15/23 11:26:44	6/15/23 11:27:43	anonymous		
13	6/15/23 11:26:23	6/15/23 11:28:03	anonymous		
14	6/15/23 11:30:41	6/15/23 11:32:01	anonymous		
15	6/15/23 11:29:32	6/15/23 11:33:43	anonymous		
16	6/15/23 11:31:02	6/15/23 11:35:57	anonymous		
17	6/15/23 11:12:04	6/15/23 11:37:54	anonymous		
18	6/15/23 11:48:19	6/15/23 11:49:11	anonymous		
19	6/15/23 11:52:23	6/15/23 11:53:30	anonymous		
20	6/15/23 12:20:54	6/15/23 12:21:20	anonymous		
21	6/15/23 12:35:17	6/15/23 12:38:23	anonymous		
22	6/15/23 12:34:20	6/15/23 12:41:18	anonymous		
23	6/15/23 12:40:12	6/15/23 12:42:08	anonymous		
24	6/15/23 12:44:01	6/15/23 12:44:52	anonymous		



(Source)

Understand data structure (CSV, text, JSON) (A2)

Text – Unstructured data:

- **Definition**: A text file is a type of digital file that contains plain text without any special formatting, such as bold, italic, or font types.
- **Characteristics of Text Files:**
  - **No Predefined Structure:** Text files lack a fixed format, making them unstructured.
  - **Challenges with Indexing:** Harder to index due to the absence of a standard structure.
  - **Difficulty in Organization:** Managing and organizing information is more complex compared to structured data formats.
  - **Irregularity:** Text files often lack regularity and decomposable internal structures, making data processing difficult.
- **Common Issues:**
  - Search and Retrieval: Without structure, searching and processing information is more challenging and may require advanced techniques like text mining or natural language processing (NLP).
- **Advantages of Text:**
  - Simplicity: Easy to create, edit, and understand without specialized software.
  - Compatibility: Universally supported across different platforms and devices.
  - Small File Size: Text files typically have smaller sizes compared to binary files, making them efficient for storing simple data.

(Source)

Understand data structure (CSV, text, JSON) (A2)

Text – Unstructured data

- Working with Text in Python:
  - Reading a Text File:

```
with open('example.txt', 'r') as file:
    data = file.read()
    print(data)
```

  - Writing to a Text File:

```
with open('example.txt', 'w') as file:
    file.write("This is a new line of text.\n")
```

  - Appending to a Text File:

```
with open('example.txt', 'a') as file:
    file.write("Appending another line of text.\n")
```

example.txt — Edited

```
Data Collection: Gathering raw data from various sources.


Data Cleaning: Removing or correcting errors, missing values, or inconsistent data.


Data Transformation: Converting data from one format or structure to another.


Data Reduction: Reducing the volume of data through aggregation or filtering while retaining
important information.


Data Analysis: Applying statistical methods and algorithms to analyze data and derive conclusions or
identify patterns.
```

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Understand data structure (CSV, text, JSON) (A2)

Text – Unstructured data:

- Scenario: we have a large collection of unformatted text data.

You need to write wrangling code in order to
  – Check if it contains any IP addresses (e.g. 128.250.65.5)
  – Find all the IP addresses

- **Requirement:**
  - Do it succinctly
  - Do it unambiguously
  - Have maintainable code

- Specifying patterns in text - **regular expressions**

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Understand data structure (CSV, text, JSON) (A2)

Text – Unstructured data:

- **Regular expressions:** Good for computing statistics, checking integrity, filtering, substitutions...

- Specifying patters in text
  - *'.' matches any character*
  - *'^' matches start of string*
  - *'$' matches end of string*
  - *'*' zero or more repetitions*
  - *'+' one or more repetitions*
  - *'|' the "or" operator, used in conjunction with parantheses ()*
  - *'[]' a set of characters, e.g. [abcd] or [a-zA-Z]*
- *https://docs.python.org/2/howto/regex.html*
- *Regex101.com*

e.g.

```
import re

text = "Contact us at support@example.com or visit our website at http://example.com."
pattern = r"\b\w+@\w+\.\w+\b"

match = re.search(pattern, text)

if match:
    print("Email found:", match.group())
```

(This code searches for an email address in the text.)

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Understand data structure (CSV, text, JSON) (A2)

Text – Unstructured data:

- **Advantages of using regular expression:**
  - Regular expressions are useful in finding, replacing and extracting information from text, such as log files..
    - Search a document for color or neighbor with or without 'u'
    - Covert a tab-delimited file to a comma-delimited file
    - Find duplicated words in a text
    - Search and replace "Bob" and "Bobby" with "Robert"

- **Regular expression are useful in verifying whether input fits into text pattern**
  - Verifying phone numbers
  - Emails
  - date

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Understand data structure (CSV, text, JSON) (A2)

More structure – JavaScript Object Notation (JSON):

- JSON is simpler and more compact/lightweight than XML. Easy to parse
- JSON example:

```
{
  "firstName": "David",
  "lastName": "Lynn",
  "isAlive": true,
  "age": 25,
  "height_cm": 167.6,
  "address": {
      "streetAddress": "211 Fox Street",
      "city": "Greenville",
      "state": "NH",
      "postalCode": "80021"
 },
```

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Why is pre-processing needed? (A2)

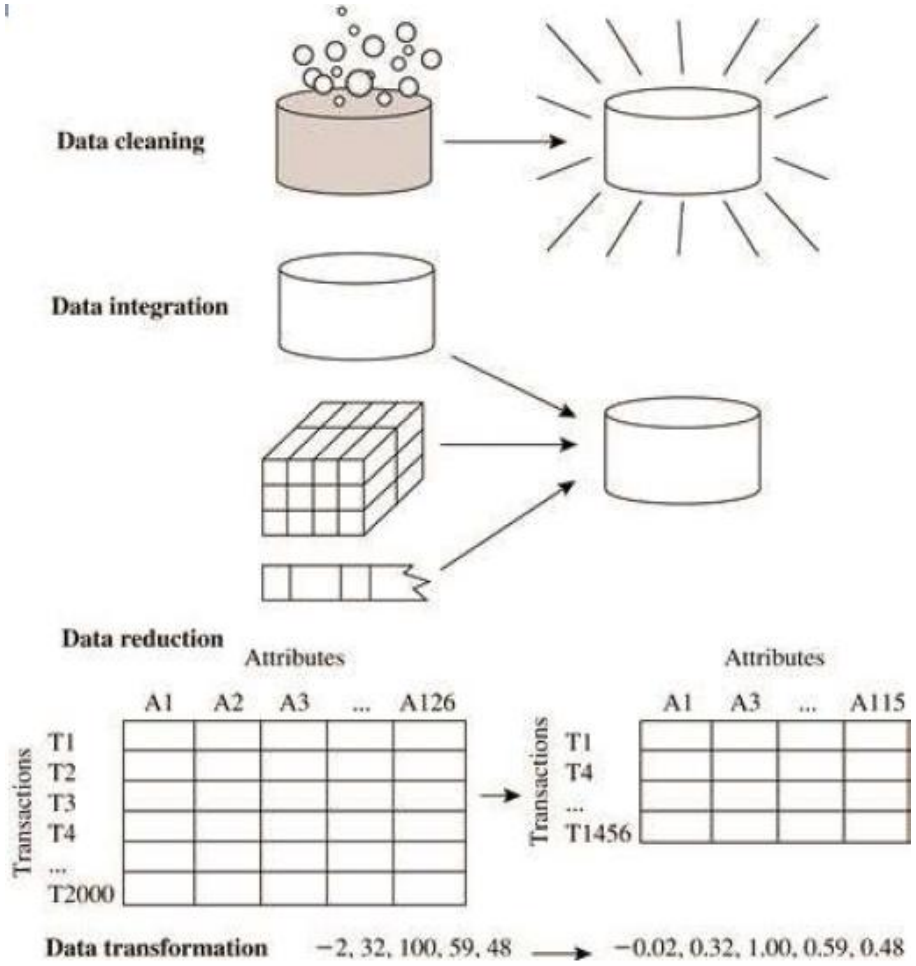
Measuring Data Quality:

- **Accuracy:**
  - **Definition:** Ensures that the data is correct and free from errors.
  - **Challenge:** Incorrect data can lead to faulty analysis or decisions.
  - **Example:** A date recorded as "15 years ago" instead of a specific date.
- **Completeness:**
  - **Definition:** Ensures that all necessary data is recorded and available.
  - **Challenge:** Missing data can skew results or make certain analyses impossible.
  - **Example:** Fields with / indicating missing date of birth.
- **Consistency:**
  - **Definition:** Ensures that data is uniform across datasets and entries.
  - **Challenge:** Inconsistent data formats can complicate processing and analysis.
  - **Example:** Different date formats like 20/11/79, 5/20/79, and 12/30/2003.
- **Timeliness:**
  - **Definition:** Ensures that the data is up-to-date and relevant.
  - **Challenge:** Outdated data may not reflect current realities and lead to incorrect conclusions.
  - **Example:** Data not updated in a timely manner.
- **Believability:**
  - **Definition:** Ensures that data is credible and trustworthy.
  - **Challenge:** Data that is hard to interpret or understand may reduce confidence in its accuracy.
  - **Example:** Mixed formats or vague entries that make it hard to assess the data.

Data of Birth
15 years ago
20/11/79
5/20/79
/
/
12/30/2003

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Major data preprocessing activities (A2)



- **Data Cleaning**: Removing or correcting errors, missing values, or inconsistent data.
- **Data integration**: Integrating multiple data

- **Data Reduction**: Reducing the volume of data through aggregation or filtering while retaining important information.
- **Data Transformation**: Converting data from one format or structure to another.

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Data cleaning - The Process (A2)

Overview of Data Cleaning Tools:

- **Popular Tools:**
  - **Google Refine (now OpenRefine):**
    - Ideal for exploring, cleaning, and transforming data.
  - **Kettle (Pentaho Data Integration):**
    - Supports ETL (Extract, Transform, Load) processes with a graphical interface.
  - **Talend:**
    - A comprehensive platform for data integration, big data, and ETL tasks.

Key Data Cleaning Tasks:

- **Data Scrubbing:**
  - **Definition:** Removing or correcting inaccurate or corrupt data.
  - **Example:** Removing erroneous entries like "N/A" or "unknown."
- **Data Discrepancy Detection:**
  - **Definition:** Identifying and resolving inconsistencies in data entries.
  - **Example:** Ensuring uniform date formats (e.g., MM/DD/YYYY).
- **Data Auditing:**
  - **Definition:** Systematic examination of data to ensure quality and consistency.
  - **Example:** Verifying that all required fields are populated.
- **ETL (Extract, Transform, Load):**
  - **Definition:** The process of extracting data from sources, transforming it into a suitable format, and loading it into a target system.
  - **Example:** Migrating data from a CSV file to a database while cleaning and formatting it.

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Data cleaning - The Process (A2)

- Our emphasis will be to understand some of the methods employed by some of these tools to deal with:
  - Noisy data
  - Inconsistent data
  - Incomplete (missing data)

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Noisy data (A2)

Examples of Noisy Data:

- **Truncated Fields:**
  - **Issue:** Data fields are cut off due to exceeding a character limit (e.g., 80 characters).
  - **Impact:** Loss of critical information and incomplete data entries.
- **Incorrectly Split Text:**
  - **Issue:** Text that should be in one cell is improperly split across multiple cells.
  - **Impact:** Misalignment of data, leading to incorrect analysis.
- **Invalid Values:**
  - **Example:** A salary recorded as "-5," which is not a valid value.
  - **Impact:** Distorts statistical analysis and data interpretation.

Common Causes of Noisy Data:

- **Imprecise Instruments:**
  - **Description:** Measurement tools that lack precision or calibration.
  - **Example:** Sensors that record data with slight deviations, leading to inaccurate readings.
- **Data Entry Issues:**
  - **Description:** Human errors during manual data entry.
  - **Example:** Typing mistakes, such as entering "52.3" instead of "5.23."
- **Data Transmission Issues:**
  - **Description:** Errors occurring during the transfer of data between systems.
  - **Example:** Corruption of data files during network transmission, leading to incomplete or garbled records.

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Inconsistent data (A2)

Examples of Inconsistent Data:

- **Different Naming Representations:**
  - **Issue:** Variations in how names or entities are represented.
  - **Example:** "Swinburne University of Technology" vs. "Swinburne" or "1" vs. "ONE".
  - **Impact:** Can cause confusion, duplicate records, or issues in data aggregation.
- **Different Date Formats:**
  - **Issue:** Inconsistent formats for dates across the dataset.
  - **Example:** "01.11.1994" (DD.MM.YYYY) vs. "01/11/1994" (MM/DD/YYYY).
  - **Impact:** Misinterpretation of dates leading to errors in analysis.
- **Duplicate Unique IDs:**
  - **Issue:** Two records with the same unique identifier.
  - **Example:** Two people with the same ID number.
  - **Impact:** Complicates data integrity and can lead to inaccuracies in record retrieval.
- **Outliers:**
  - **Example:** Age of students recorded as {22, 17, 23, 33, 24, 34, 999}.
  - **Issue:** 999 is clearly an outlier and not a realistic age.
  - **Impact:** Skews statistical analysis and may indicate an error in data entry or transmission.

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Inconsistent data (A2)

Handling Inconsistent Data:

•For Outliers:

- **Condition 1: Keep It**
  - **When:** The outlier is reasonable and within acceptable limits.
- **Condition 2: Impute It**
  - **When:** Domain knowledge is insufficient to accurately determine the correct value.
  - **Action:** Use statistical methods or average values for imputation.
- **Condition 3: Correct It**
  - **When:** Domain knowledge allows for accurate inference.
  - **Example:** Replace 999 with a realistic age based on the context.

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Missing or incomplete data (A2)

Lacking Feature Values:

•Common Scenarios:

- **Empty Fields:**
  - **Example:** Name = ” " (empty string)
- **Null Values:**
  - **Example:** Age = null (no data recorded)
- **Impact:** Missing values can lead to incomplete analysis, bias, and errors in data interpretation.

Types of Missing Data (Rubin, 1976):

•Missing Completely at Random (MCAR):

- **Definition:** Data is missing independently of both observed and unobserved data. The missingness has no pattern or relationship with any other data.
- **Example:** Flipping a coin to decide whether or not to answer an exam question.
- **Impact:** MCAR data can be handled more easily as it doesn’t introduce bias.

•Missing Not Completely at Random (MNAR):

- **Definition:** The likelihood of data being missing is related to the unobserved data. There is a pattern or reason behind the missingness.
- **Example:** A survey on health, where non-responders might be less healthy, leading to non-random missing data.
- **Impact:** MNAR data can introduce bias and requires more sophisticated handling techniques.

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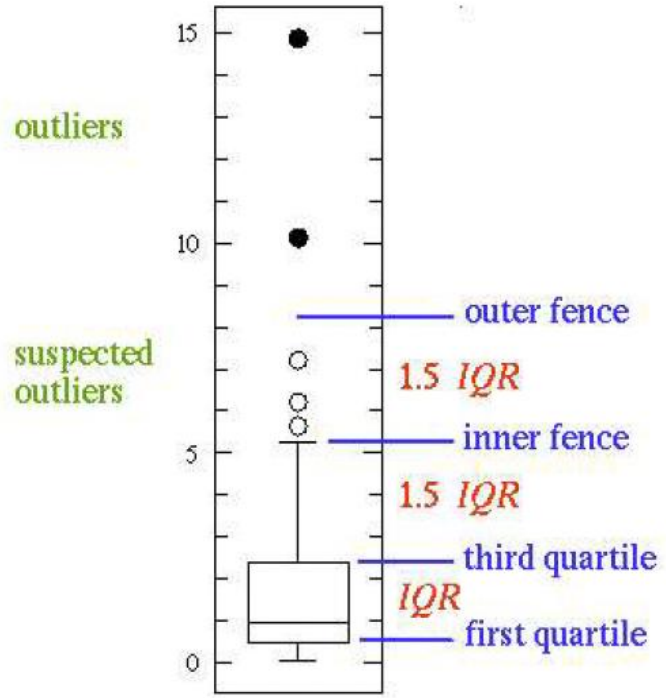
Dealing with missing data (A2)

- Delete all instances with a missing value
  - Sometimes called case deletion
  - Effects
    - Easy to analyse the new
    - May produce bias on analysis if new sample size small or structure exists in the missing data
- Manually correct
  - A human eyeballs the missing value and fills it in using their expert knowledge
- Imputation: imputing all missing values (e.g., fill in with zeros, means and median...)

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Outlier Analysis (A2)

- Outlier: A data object that deviates significantly from the normal objects as if it were generated by a different mechanism
- From a statistics perspective:
  - Suspected outliers :
    - >1.5 IQR above third quartile
    - >1.5 IQR below 1st quartile
- How to deal with outlier ?
  - Similar to missing data



The **Interquartile Range (IQR)** is a measure of statistical dispersion, which represents the range within which the middle 50% of the data points lie. It is calculated by subtracting the first quartile (Q1) from the third quartile (Q3) in a dataset.

Key Points about IQR:

•**Quartiles:**

- **First Quartile (Q1):** The median of the lower half of the dataset, which represents the 25th percentile.
- **Third Quartile (Q3):** The median of the upper half of the dataset, which represents the 75th percentile.

•**Calculation:**

•IQR=Q3−Q1

Thank you

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